

1. Administrative & Study Identification

Full Study Title A Phase I/II Open-label, Dose Escalation and Dose Expansion Study to Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, and Efficacy of AZD6621, a T Cell-engaging Antibody That Targets STEAP2, CD3, and CD8 in Adult Participants With Metastatic Prostate Cancer **Short Title/Acronym** ACTIVATED-4-PC **Protocol Number** D8020C00003 **NCT Number** NCT07192614 **Sponsor Name** AstraZeneca **Investigational Product** AZD6621 (T Cell-engaging Antibody Targeting STEAP2, CD3, CD8) **Phase of Development** Phase I/II **Medical Monitor** [AstraZeneca Clinical Operations / Medical Monitor]

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the safety, tolerability, pharmacokinetics, pharmacodynamics, and efficacy of AZD6621 in adult participants with Metastatic Castration-Resistant Prostate Cancer (mCRPC) (Part A: dose escalation; Part B: dose expansion). * **Secondary Objectives:** To characterize the specific anti-tumor activity (e.g., PSA response, objective response rate) and the immunogenicity of AZD6621 over time.

2.2 Background & Rationale AZD6621 is a novel, first-in-human T cell-engaging antibody designed to activate the body's immune system (T cells) to attack advanced prostate cancer cells. This study is necessary to determine the optimal dose that balances efficacy with minimal side effects for patients with Metastatic Castration-Resistant Prostate Cancer (mCRPC) who have progressed despite ongoing androgen deprivation therapy.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm (Dose escalation and expansion) * **Blinding:** Open-label * **Randomization:** Sequential Assignment / Randomized (depending on module)

3.2 Planned Enrollment & Duration * **Target N\$:** 52 participants * **Number of Sites:** Multicenter (including sites in the USA, South Korea, China, Japan, UK, Belgium, and Netherlands) * **Estimated Study Start:** 30-Sept-2025 * **Estimated Primary Completion:** 29-Mar-2029 * **Total Duration:** ~42 months (Estimated Primary Completion: Mar 2029)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male, ≥ 18 years of age at the time of signing the informed consent. 2. Histologically confirmed

diagnosis of metastatic adenocarcinoma of the prostate. 3. Surgically or medically castrated, with serum testosterone levels ≤ 50 ng/dL (≤ 1.75 nmol/L); participants must be currently taking and willing to continue LHRH agonist or antagonist therapy. 4. Evidence of disease progression within 6 months (e.g., minimum of 3 rising PSA levels with PSA ≥ 1 ng/mL, or radiographic progression).

4.2 Exclusion Criteria 1. Previous CAR-T cell therapy within the last 6 months. 2. History of Grade ≥ 3 Cytokine Release Syndrome (CRS) or Grade ≥ 2 Immune effector cell-associated neurotoxicity syndrome (ICANS) with prior therapy. 3. Unresolved toxicity from prior anti-cancer therapy of CTCAE Grade ≥ 2 . 4. Inadequate cardiac function (LVEF $< 50\%$) or QT prolongation (QTc > 470 ms). 5. Active or prior documented autoimmune/inflammatory disorders requiring permanent immunosuppressive therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Management Pathway:** Dose Escalation (Part A) followed by Dose Expansion (Part B). * **Dosage/Frequency:** Escalating doses of AZD6621 to identify the maximum tolerated dose (Part A), followed by target doses in a larger cohort (Part B). * **Route of Administration:** Intravenous (IV) or Subcutaneous (SC) injection. * **Checklist Review Interval:** Prior to each dose administration. * **Duration of Treatment:** Until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy * **Permitted:** LHRH agonist or antagonist (mandatory maintenance). * **Prohibited:** Live attenuated vaccine within 30 days prior to the first dose, and COVID-19 vaccination within 72 hours prior.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, PSA/Testosterone Labs, Baseline Tumor Biopsy
Baseline	Day 0	Treatment allocation, First Dose, Safety Labs
Interim	Per Cycle	PK/PD assessments, DLT evaluation, CRS/ICANS monitoring
End of Study	Follow-up	Final Vital Signs, RECIST/PCWG3 scans, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Incidence of Adverse Events and Dose Limiting Toxicities (DLTs) to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose determination. * **Measurement Method:** Clinical grading via CTCAE v5.0 and ASTCT criteria for CRS/ICANS.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetics (Cmax, AUC) of AZD6621. * Anti-tumor activity (Objective Response Rate, PSA decline/response). * Immunogenicity (Presence of Anti-Drug Antibodies).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Intensive inpatient/outpatient monitoring for Dose Limiting Toxicities, CRS, and ICANS.
- **Onsite Visit Cadence:** Intensive during Screening and initial Treatment Cycles.
- **Serious Adverse Events (SAEs):** Standard 24-hour reporting via investigator documentation.
- **Data Monitoring Committee (DMC):** Safety Review Committee (SRC) oversight to guide dose escalation thresholds.
- **Quality Audit Mechanism:** Safety Review Committee (SRC) oversight for dose escalation decisions.

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all participants receiving \$1\$ dose of AZD6621).
- **Sample Size Justification:** \$N=52\$; powered for standard modular dose-escalation (Part A) and dose-expansion (Part B) modeling.
- **Handling of Missing Data:** Last Observation Carried Forward (LOCF) for efficacy evaluations; safety data observed as treated.

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** High-frequency onsite monitoring during dose-escalation cohorts.
- **Remote Monitoring Frequency:** Between clinical visits per Phase I/II guidelines.
- **Audit Trail:** eCRF locking and comprehensive data clarification forms.
- **Education Protocol (HCP):** Investigator Site Training for CRS/ICANS Management and PK sampling.
- **Education Protocol (Patient):** Onsite Informed Consent & Dose Administration Safety Briefing.

11. Study Identifiers & Governance

Primary ID: {{D8020C00003}} **Registry Number:** {{NCT07192614}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{ACTIVATED-4-PC}} **Official Regulatory Title:** {{A Phase I/II Open-label, Dose Escalation and Dose Expansion Study to Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, and Efficacy of

AZD6621, a T Cell-engaging Antibody That Targets STEAP2, CD3, and CD8 in Adult Participants With Metastatic Prostate Cancer}}

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: {{Metastatic Castration-Resistant Prostate Cancer (mCRPC)}} Diagnosis Threshold: {{PSA $\geq$ 1\$ ng/mL, Testosterone $\leq$ 50\$ ng/dL}} Medical Methodology: {{T Cell-engaging Antibody (Targeting STEAP2, CD3, CD8)}} Optimization Target: {{Maximum Tolerated Dose / Recommended Phase 2 Dose determination}}

13. Study Procedures & Methodology

Management Pathway: {{Dose Escalation (Part A) followed by Dose Expansion (Part B)}} Education Protocol (HCP): {{Investigator Site Training for CRS/ICANS Management and PK sampling}} Education Protocol (Patient): {{Onsite Informed Consent & Dose Administration Safety Briefing}} Quality Audit Mechanism: {{Safety Review Committee (SRC) oversight for dose escalation decisions}}

14. Observation & Follow-Up Schedule

Total Duration: {{~42 months (Estimated Primary Completion: Mar 2029)}} Onsite Visit Cadence: {{Intensive during Screening and initial Treatment Cycles}} Remote Monitoring Frequency: {{Between clinical visits per Phase I/II guidelines}} Checklist Review Interval: {{Prior to each dose administration}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					